

Cell Design Specification — VBF-T5R3-DS-01

Case: t5_r3 — next-generation flagship vehicle battery · **Generated:** 2026-08-26 · **Status:** DFN-verified concept (virtual test), true-compute endorsement skipped (real_compute=false)

1. Basic Specification

Item	Value	Source
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Electrochemical system	NMC811 / graphite (Chen2020 baseline parameterization)	`param_dump.txt` Chen2020; base determined in funnel entry (start_stage=3)
Nominal capacity	5.0 Ah (parameter set nominal)	`param_dump.txt`: Nominal cell capacity [A.h]
Verified 1C capacity	5.0442 Ah (full-order DFN)	`cell/r6_y4_1c_dfn.json:capacity_ah`
Voltage window	2.5 – 4.2 V	`param_dump.txt`: Upper/Lower voltage cut-off
Midpoint voltage	3.832 V (1C discharge)	`cell/r6_y4_energy_dfn.json:midpoint_voltage_v`
Cell dimensions	Stack thickness 178.8 μm × electrode area 0.1027 m² (height 0.065 m × width 1.58 m)	`r6_y4_energy_dfn.json:thickness_m/area_m2`; geometry `param_dump.txt`
Shell / enclosure thickness	Not provided (no parameter)	—
Electrolyte formulation	Chen2020 baseline (LiPF6 EC/EMC-type) with transport upgrade: conductivity 5.0 S/m (constant), cation transference number 0.6 , diffusivity 9e-10 m²/s	`cell/params_y4.json` (design overrides); no additive candidates invented (case starts at Stage 3 — recorded in funnel entry)
Cation transference number	0.6 (baseline default 0.2594)	`cell/params_y4.json`

2. Electrode and Separator

Layer	Thickness	Porosity	AM volume fraction	Density	Source
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Positive electrode (NMC811)	75.6 μm	0.335	0.665	3262 kg/m³	`param_dump.txt` Chen2020 (baseline, not overridden)
Negative electrode (graphite)	85.2 μm	0.25	0.75	1657 kg/m³	`param_dump.txt` Chen2020
Separator	7 μm (baseline 12 μm)	0.47	—	(mass ref 210 kg/m³ eq.)	`cell/params_y4.json` override + `param_dump.txt`
Positive current collector	Al 6 μm (baseline 16 μm)	—	—	2700 kg/m³	`cell/params_y4.json` + `param_dump.txt`
Negative current collector	Cu 5 μm (baseline 12 μm)	—	—	8960 kg/m³	`cell/params_y4.json` + `param_dump.txt`
Particle radii	Positive + negative both 3 μm (baseline 5.22 / 5.86 μm)	—	—	—	`cell/params_y4.json`

N/P ratio ~ 1.19 — areal capacities: negative = 8.52e-4 cm × 0.75 × 1.657 g/cm³ × 372 mAh/g = 3.94 mAh/cm²; positive = 7.56e-4 × 0.665 × 3.262 × 201 = 3.30 mAh/cm². *(Specific capacities 372 / 201 mAh/g are domain-standard literature defaults, annotated — Chen2020 parameter set does not expose per-electrode capacity-density keys.)*

3. Process Design Parameters

Parameter	Value	Formula	Note
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Positive areal density	163.99 g/m²	thickness × (1-ε) × ρ = 16.842 g ÷ 0.1027 m²	mechanical
Negative areal density	105.88 g/m²	10.874 g ÷ 0.1027 m²	mechanical
Positive compaction density	2.169 g/cm³	3262 × 0.665 ÷ 1000	±1000 note per spec
Negative compaction density	1.243 g/cm³	1657 × 0.75 ÷ 1000	±1000 note per spec

Parameter	Value	Formula	Note
Electrolyte fill amount	6.15 g	pore volume $5.126 \text{ cm}^3 \times 1.2 \text{ g/cm}^3$ (fill factor 1.0)	electrolyte density = literature value 1.2 g/cm^3 , annotated
Formation recommendation	0.1C CC to 4.2 V, 25 °C, 2 cycles	—	design recommended value; actual production-line value requires tuning

4. Mass Breakdown (DFN-verified geometry)

Layer	Mass (g)	Basis
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Positive electrode (coating)	16.842	`r6_y4_energy_dfn.json:layer_kg_m2.positive_electrode` × 0.1027 m²
Negative electrode (coating)	10.874	same, negative_electrode
Positive current collector (Al)	1.664	layer_kg_m2.positive_cc
Negative current collector (Cu)	4.601	layer_kg_m2.negative_cc
Separator	0.151	layer_kg_m2.separator
Total (contract caliber, electrolyte excluded)	**34.132**	`r6_y4_energy_dfn.json:mass_kg` — matches Σ automatically
Electrolyte (est., annotated)	6.15	pore volume × 1.2 g/cm^3
Total incl. electrolyte	40.28	contract formula excludes electrolyte — noted

5. Performance Verification (thresholds from task text, entry-0 contract)

Item	Value	Threshold	Determination	Source
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1C capacity	5.0442 Ah	no contractual threshold (info)	—	`r6_y4_1c_dfn.json:capacity_ah`
Energy density	**535.72 Wh/kg**	$\geq 500.94 \text{ Wh/kg}$	** (PASS) PASS **	`r6_y4_energy_dfn.json:energy_density_wh_kg` (contract formula, electrolyte excluded)
Volumetric energy density	995.79 Wh/L	info	—	`r6_y4_energy_dfn.json:energy_density_wh_l`
4C max temperature	**327.60 K (54.4 °C)**	$\leq 333.15 \text{ K}$ ($\leq 60 \text{ °C}$)	** (PASS) PASS **	`r6_y4_4c_dfn.json:T_max_K`
4C lithium plating	min anode potential **+0.0200 V** (never < 0)	plated = false	** (PASS) PASS **	`r6_y4_4c_dfn.json:anode_potential_v`
4C CC charge window	0 -> ~69% SOC (3.48 Ah) in 626 s at 20 A, then 4.2 V cut-off	info (task = "support 4C fast charge (no plating)"; CV tail at decaying current, plating risk monotonic with current — domain reasoning, not simulated)	—	`r6_y4_4c_dfn.json:time_s/voltage_v` (charge step, 20 A)
DC resistance	2.723 mΩ	info	—	`r6_y4_energy_dfn.json:dcr_ohm`
Power density	45,030 W/kg	info	—	`r6_y4_energy_dfn.json:power_density_w_kg`

Verified with full-order DFN (`model\_used: DFN`, no solver fallback).

6. Design Notes

- **Why these changes****: baseline Chen2020 = ED 400.29 Wh/kg / 4C plated (min_ap −0.438 V, CC window ~26 s) / T_max 332.35 K (round 1). Root cause of plating = electrolyte salt transport (4 isolation simulations, round 2). Levers: thin CC/sep/particles + transport upgrade (σ 5.0 S/m, t_{sep} 0.6, D_e 9e-10) + double-sided cooling area 0.01062 m² + mild thermal tuning $h=26 \text{ W/m}^2\text{K}$ (~6.6 mV/K kinetic margin leverages self-heating; finalists warm->327.6–331.2 K, still $\geq 2.5 \text{ K}$ below 333.15 K). Honest true-4C rebuild at 5 Ah actual (round 4 honesty probes caught an earlier 6.55 Ah nameplate artifact and it was rejected). Finalists Y4 (above) and Y3 (Al 8 μm / sep 9 μm, ED 526.24, min_ap +0.0162 V) both DFN-verified (round

6).

- 5 μm Cu / 6 μm Al foils are aggressive for winding — Y3 (8 μm Al / 9 μm sep) is the supply-friendly alternate with almost identical electrical performance.
- Full rationale per round: `log.jsonl` rounds 1–6 (propose/evaluate entries).